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Systems and methods for performing inpainting

Abstract

Described embodiments generally relate to a computer-implemented method for performing inpainting. The method includes accessing a first image; receiving a selected area of the first image; identifying a foreground area of the first image; generating a merged mask based on the union of the user selected area and the foreground area; performing an inpainting process on the area of the first image corresponding to the merged mask to generate a second image, being an inpainted image; generating a reduced mask based on the user selected area reduced by the foreground area; and combining the first image with the area of the second image corresponding to the reduced mask to produce an output image.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS
(1) This application is a U.S. Non-Provisional Application that claims priority to and the benefit of Australian Patent Application No. 2023204097, filed Jun. 27, 2023, that in turn claims priority to Australian Provisional Patent Application No. 2023901771, filed Jun. 5, 2023, that are hereby incorporated by reference in their entirety.
TECHNICAL FIELD
(2) Described embodiments relate to systems and methods for performing inpainting. In particular, described embodiments relate to systems and methods for performing inpainting of digital images.
BACKGROUND
(3) Inpainting is a term used to describe an image processing technique for removing unwanted image elements, and replacing these with new elements in a way that preserves the realism of the original image. Historically, inpainting was performed manually to physical images by painting or otherwise covering unwanted image elements with a physical medium. As digital image processing tools became widely adopted, digital inpainting became possible.
(4) Digital inpainting can be performed manually using digital image editing software to fill in areas of an image. This can be extremely long and tedious work if a quality result is desired, especially when working with large areas. This is because this method of inpainting can require a pixel-level manipulation of the image to replace the unwanted area with a realistic fill. Some automated approaches have been developed, but these often produce an unrealistic result and undesirably distort the image.
(5) It is desired to address or ameliorate one or more shortcomings or disadvantages associated with prior systems and methods for performing inpainting, or to at least provide a useful alternative thereto.
(6) Throughout this specification the word “comprise”, or variations such as “comprises” or “comprising”, will be understood to imply the inclusion of a stated element, integer or step, or group of elements, integers or steps, but not the exclusion of any other element, integer or step, or

group of elements, integers or steps.

(7) Any discussion of documents, acts, materials, devices, articles or the like which has been included in the present specification is not to be taken as an admission that any or all of these matters form part of the prior art base or were common general knowledge in the field relevant to the present disclosure as it existed before the priority date of each of the appended claims.

SUMMARY

(8) Some embodiments relate to a computer-implemented method for performing inpainting, the method comprising: accessing a first image; receiving a selected area of the first image; identifying a foreground area of the first image; generating a merged mask based on the union of the user selected area and the foreground area; performing an inpainting process on the area of the first image corresponding to the merged mask to generate a second image, being an inpainted image; generating a reduced mask based on the user selected area reduced by the foreground area; and combining the first image with the area of the second image corresponding to the reduced mask to produce an output image.

(9) Some embodiments further comprise storing the output image to a memory location.

(10) Some embodiments further comprise communicating the output image to an external device.

(11) Some embodiments further comprise: computing a relative mask overlap between the user selected area and the foreground area; comparing the relative mask overlap with at least one threshold; and proceeding with the steps of generating a merged mask, performing an inpainting process, generating a reduced mask and combining images based on the results of the comparison, wherein these steps form part of a foreground protection process.

(12) According to some embodiments, the relative mask overlap is calculated using the equation:

(13) $RMO = \frac{\text{Num\_pixels(US\_area .Math. FG\_area)}}{\text{Num\_pixels(US\_area)}}$

where RMO is the relative mask overlap, Num_pixels is a function to find a number of pixels, US_area is the user selected area and FG_area is the foreground area.

(14) According to some embodiments, comparing the relative mask overlap with at least one threshold comprises comparing the relative mask overlap with a lower threshold, and performing the foreground protection process based on determining that the relative mask overlap is greater than the lower threshold.

(15) In some embodiments, the lower threshold is 0%.

(16) In some embodiments, based on determining that the relative mask overlap is less than or equal to the lower threshold, instead of performing the foreground protection algorithm, performing an inpainting process on the area of the first image corresponding to the user selected area to generate the output image.

(17) According to some embodiments, comparing the relative mask overlap with at least one threshold comprises comparing the relative mask overlap with an upper threshold, and performing the foreground protection process based on determining that the relative mask overlap is less than the upper threshold.

(18) In some embodiments, the upper threshold is between 10% and 90%.

(19) In some embodiments, the upper threshold is between 40% and 70%.

(20) In some embodiments, the upper threshold is 50%.

(21) According to some embodiments, based on determining that the relative mask overlap is more than or equal to the upper threshold, instead of performing the foreground protection process, performing an inpainting process on the area of the first image corresponding to the user selected area to generate the output image.

(22) Some embodiments further comprise the step of dilating the merged mask, and performing the inpainting process on the area of the first image corresponding to the dilated merged mask.

(23) In some embodiments, the merged mask is dilated by between 1 and 9 pixels.

(24) In some embodiments, the merged mask is dilated by between 3 and 7 pixels.

(25) In some embodiments, the merged mask is dilated by 5 pixels.

(26) Some embodiments further comprise performing a second inpainting process on the area of the first image corresponding to the user selected area to generate the second image.
(27) Some embodiments further comprise performing a smoothing process on the output image before outputting the image, to smooth the edge between the first image and the second image.
(28) According to some embodiments, the smoothing is performed using an alpha compositing process.
(29) In some embodiments, performing the inpainting process comprises performing an AI based inpainting process.
(30) According to some embodiments, performing the inpainting process comprises performing a LaMa inpainting process.
(31) In some embodiments, identifying a foreground area of the first image comprises performing a background removal process on the first image.
(32) Some embodiments relate to a non-transitory computer-readable storage medium storing instructions which, when executed by a processing device, cause the processing device to perform the method of some other described embodiments.
(33) Some embodiments relate to a computing device comprising: the non-transitory computer-readable storage medium of some other described embodiments; and a processor configured to execute the instructions stored in the non-transitory computer-readable storage medium.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) Various ones of the appended drawings merely illustrate example embodiments of the present disclosure and cannot be considered as limiting its scope.
(2) FIGS. **1**A to **1**D show a first series of example images that show the results of inpainting using prior known techniques compared to methods of some embodiments described below;
(3) FIGS. **2**A to **2**D show a second series of example images that show the results of inpainting using prior known techniques compared to methods of some embodiments described below:
(4) FIG. **3** is a schematic diagram of a system for performing inpainting according to some embodiments;
(5) FIG. **4** is a process flow diagram of a method of performing inpainting according to some embodiments;
(6) FIGS. **5**A to **5**K are example diagrams showing the steps of performing the method of FIG. **4** according to some embodiments;
(7) FIGS. **6**A to **6**C show a series of example images showing the results of performing a background separation step according to some embodiments;
(8) FIGS. **7**A to **7**D show a series of example images showing the results of performing a relative mask overlap calculation step according to some embodiments;
(9) FIGS. **8**A to **8**D show a series of example images showing the results of performing a merged mask generation step according to some embodiments;
(10) FIGS. **9**A to **9**C show a series of example images showing the results of performing a merged mask inpainting step according to some embodiments;
(11) FIGS. **10**A to **10**C show a series of example images showing the results of performing a user selected area inpainting step according to some embodiments;
(12) FIGS. **11**A to **11**D show a series of example images showing the results of performing a reduced mask generation step according to some embodiments;
(13) FIGS. **12**A to **12**D show a series of example images showing the results of performing a reduced mask inpainting step according to some embodiments; and
(14) FIGS. **13**A to **13**D show a series of example images showing the results of performing a

smooth image merging step according to some embodiments.

DESCRIPTION OF EMBODIMENTS

(15) Described embodiments relate to systems and methods for performing inpainting. In particular, described embodiments relate to systems and methods for performing inpainting of digital images.

(16) Image inpainting refers to removing an area of an image, such as an undesirable image element, and filling the removed area such that the image looks realistic and as if the undesirable image element had never been there. Automated digital techniques have been developed that allow users to select one or more areas of an image for removal, and which will automatically fill the selected areas in a way that replaces the area with a realistic alternative. However, these techniques often distort surrounding image areas, especially when the area selected for removal overlaps with a foreground element of the image.

(17) FIGS. **1**A to **1**C and **2**A to **2**C show examples of the distortion that can occur using existing techniques.

(18) FIG. **1**A shows an example image **100** having a foreground image element **105** and a background image element **110**. In the illustrated example, foreground image element **105** comprises a person located close to the camera capturing the image and being the main subjects of the image, while background image element **110** comprises a train, people and cityscape located behind the person, serving as a backdrop. Image **100** also comprises an unwanted image element **115**, being a person who is not an intended subject of the image, but who has been partially captured in the image behind the subject of the image. The unwanted image element **115** is directly adjacent to and in contact with the edge of part of the foreground image element **105**.

(19) FIG. **1**B shows an image **120**. Image **120** has the same foreground image element **105** and background image element **110** as image **100**. The difference between image **100** and image **120** is that image **120** includes a user selected area **125** applied over unwanted image element **115**. According to some embodiments, a user wishing to edit image **100** may interact with a user interface component of a computing device displaying image **100** to indicate which area of image **100** they consider to be an unwanted image element, and wish to remove from the image. For example, the user may use a brush tool, trace tool, or other tool to digitally select, trace, circle, or “paint” over the unwanted image element via the user interface to produce a user selected area **125**, which can be used as a mask to perform an inpainting method. However, the user may inadvertently select areas that they do not want removed from image **100** as part of the user selected area **125**. For example, the user may inadvertently select areas of the foreground image element **105**. In the illustrated embodiment, user selected area **125** overlaps with a portion of the foreground image element **105** corresponding to the back and neck of the subject.

(20) FIG. **1**C shows an image **140**. Image **140** has the same foreground image element **105** and background image element **110** as image **100**. The difference between image **100** and image **140** is that image **140** has had an inpainting process performed on the area selected by the user, that is the user selected area **125**. In image **140**, the inpainting has been performed by a previously known inpainting technique, using an algorithm to fill the region indicated by user selected area **125** such that the image looks as if the unwanted image element had never been there. For example, this may be done using a machine learning or AI based inpainting method, such as Resolution-robust Large Mask Inpainting with Fourier Convolutions (LaMa), co-modulated Generative Adversarial Networks (CoModGANs), or Guided Language-to-Image Diffusion for Generation and Editing (GLIDE).

(21) In the illustrated example, the inpainting technique has removed unwanted image element **115**, but has distorted the foreground image element **105** to produce a distorted foreground portion **150**. As shown in the illustrated example, the neck, shoulder and back of the subject who forms foreground image element **105** has been blurred and distorted in the area indicated as distorted foreground portion **150**. The technique has also distorted background image element **110** to

produce a distorted background portion **145**. As shown in the illustrated example, the colour of the distorted background portion **145** does not match the surrounding areas of the background image element **110**, as the foreground image element **105** has bled into the background image element **110**. These effects may be increased the more that user selected area **125** overlaps foreground image portion **105**. The intent of the user would most likely be to leave foreground image element **105** unchanged, and to have a more realistic fill of the background image portion **110**.
(22) FIG. **2**A shows another example image **200** having a foreground image element **205** and a background image element **210**. In the illustrated example, foreground image element **205** comprises a person located close to the camera capturing the image and being the main subject of the image, while background image element **210** comprises the walls of the room that the person is located in, serving as a backdrop. Image **200** also comprises an unwanted image element **215**, being a number of picture frames, which have been partially captured in the image behind the subject of the image. The unwanted image element **215** is directly adjacent to and in contact with the edge of part of the foreground image element **205**.
(23) FIG. **2**B shows an image **220**. Image **220** has the same foreground image element **205** and background image element **210** as image **200**. The difference between image **200** and image **220** is that image **220** includes a user selected area **225** applied over unwanted image element **215**. According to some embodiments, a user wishing to edit image **200** may interact with a user interface component of a computing device displaying image **200** to indicate which area of image **200** they consider to be an unwanted image element, and wish to remove from the image. For example, the user may use a brush tool, trace tool, or other tool to digitally select, trace, circle, or “paint” over the unwanted image element via the user interface to produce a user selected area **225**, which can be used as a mask to perform an inpainting method. However, the user may inadvertently select areas that they do not want removed from image **200** as part of the user selected area **225**. For example, the user may inadvertently select areas of the foreground image element **205**. In the illustrated embodiment, user selected area **225** overlaps with a portion of the foreground image element **205** corresponding to the hair and face of the subject.
(24) FIG. **2**C shows an image **240**. Image **240** has the same foreground image element **205** and background image element **210** as image **200**. The difference between image **200** and image **240** is that image **240** has had an inpainting process performed on the area of the image selected by user selected area **225**. In image **240**, the inpainting has been performed by a previously known inpainting technique, using an algorithm to fill the region indicated by user selected area **225** such that the image looks as if the unwanted image element had never been there. For example, this may be done using a machine learning or AI based inpainting method, such as LaMa, CoModGANs or GLIDE.
(25) In the illustrated example, the inpainting technique has removed unwanted image element **215**, but has distorted the foreground image element **205** to produce a distorted foreground portion **250**. As shown in the illustrated example, the face and hair of the subject who forms foreground image element **205** has been blurred and distorted in the area indicated as distorted foreground portion **250**. The technique has also distorted background image element **210** to produce a distorted background portion **245**. As shown in the illustrated example, the background image element **210** has blurred and distorted in the area indicated as distorted background portion **245**, and the colour of foreground image element **205** has bled into distorted background portion **245**. These effects may be increased the more that user selected area **225** overlaps foreground image portion **205**. The intent of the user would most likely be to leave foreground image element **205** unchanged, and to have a more realistic fill of the background image portion **210**.
(26) The examples shown in FIGS. **1**A to **1**C and **2**A to **2**C illustrate some limitations with known methods of inpainting. Where an unwanted image element overlaps, touches, or is proximate to a foreground image element, the area selected by a user for inpainting may touch or overlap the foreground image element, causing undesirable effects. As the user needs to brush or trace close to

the foreground image element in order to completely cover or select the unwanted image element, these undesirable effects limit the usability of known inpainting tools.

(27) Described embodiments relate to a method of inpainting that preserves background and foreground image elements, to avoid the distortion shown in FIGS. **1**C and **2**C, even when areas of these image elements are selected. FIGS. **1**D and **2**D show the results of performing inpainting according to some described embodiments. As seen from these images, the unwanted image elements **115** and **215** have been removed from area **165** and **265**, respectively, leaving foreground image elements **105** and **205** and background image elements **110** and **210** undistorted and preserved, producing a more realistic effect compared to the images shown in FIGS. **1**C and **2**C.

(28) FIG. **3** is a schematic diagram showing an example system **300** that may be used for performing inpainting methods according to some described embodiments.

(29) System **300** comprises a user computing device **310** which may be used by a user wishing to edit one or more images. Specifically, user computing device **310** may be used by a user to perform inpainting on one or more images using methods as described below. In the illustrated embodiments, system **300** further comprises a server system **320**. User computing device **310** may be in communication with server system **320** via a network **330**. However, in some embodiments, user computing device **310** may be configured to perform the described methods independently, without access to a network **330** or server system **320**.

(30) User computing device **310** may be a computing device such as a personal computer, laptop computer, desktop computer, tablet, or smart phone, for example. User computing device **310** comprises a processor **311** configured to read and execute program code. Processor **311** may include one or more data processors for executing instructions, and may include one or more of a microprocessor, microcontroller-based platform, a suitable integrated circuit, and one or more application-specific integrated circuits (ASIC's).

(31) User computing device **310** further comprises at least one memory **312**. Memory **312** may include one or more memory storage locations which may include volatile and non-volatile memory, and may be in the form of ROM, RAM, flash or other memory types. Memory **312** may also comprise system memory, such as a BIOS.

(32) Memory **312** is arranged to be accessible to processor **311**, and to store data that can be read and written to by processor **311**. Memory **312** may also contain program code **314** that is executable by processor **311**, to cause processor **311** to perform various functions. For example, program code **314** may include an image editing application **315**. Processor **321** executing image editing application **315** may be caused to perform inpainting methods, as described below in further detail with reference to FIGS. **4** to **13**D.

(33) According to some embodiments, image editing application **315** may be a standalone application in the form of a smartphone application or desktop application that provides image editing functionality. In some embodiments, image editing application **315** may be a web browser application (such as Chrome, Safari, Internet Explorer, Opera, or an alternative web browser application) which may be configured to access web pages that provide image editing functionality via an appropriate uniform resource locator (URL).

(34) Program code **314** may include additional applications that are not illustrated in FIG. **3**, such as an operating system application, which may be a mobile operating system if user computing device **310** is a mobile device, a desktop operating system if user computing device **310** is a desktop device, or an alternative operating system.

(35) User computing device **310** may further comprise user input and output peripherals **316**. These may include one or more of a display screen, touch screen display, mouse, keyboard, speaker, microphone, and camera, for example. User I/O **316** may be used to receive data and instructions from a user, and to communicate information to a user.

(36) User computing device **310** also comprises a communications interface **317**, to facilitate communication between user computing device **310** and other remote or external devices.

Communications module **317** may allow for wired or wireless communication between user computing device **310** and external devices, and may utilise Wi-Fi, USB, Bluetooth, or other communications protocols. According to some embodiments, communications module **317** may facilitate communication between user computing device **310** and server system **320**, for example.

(37) Network **330** may comprise one or more local area networks or wide area networks that facilitate communication between elements of system **300**. For example, according to some embodiments, network **330** may be the internet. However, network **330** may comprise at least a portion of any one or more networks having one or more nodes that transmit, receive, forward, generate, buffer, store, route, switch, process, or a combination thereof, etc. one or more messages, packets, signals, some combination thereof, or so forth. Network **330** may include, for example, one or more of: a wireless network, a wired network, an internet, an intranet, a public network, a packet-switched network, a circuit-switched network, an ad hoc network, an infrastructure network, a public-switched telephone network (PSTN), a cable network, a cellular network, a satellite network, a fibre-optic network, or some combination thereof.

(38) Server system **320** may comprise one or more computing devices and/or server devices, such as one or more servers, databases, and/or processing devices in communication over a network, with the computing devices hosting one or more application programs, libraries, APIs or other software elements. The components of server system **300** may provide server-side functionality to one or more client applications, such as image editing application **315**. The server-side functionality may include operations such as user account management, login, and content creation functions such as image editing, saving, publishing, and sharing functions. According to some embodiments, server system **320** may comprise a cloud based server system. While a single server system **320** is shown, server system **320** may comprise multiple systems of servers, databases, and/or processing devices. Server system **320** may host one or more components of a platform for performing inpainting according to some described embodiments.

(39) Server system **320** may comprise at least one processor **321** and a memory **322**. Processor **321** may include one or more data processors for executing instructions, and may include one or more of a microprocessor, microcontroller-based platform, a suitable integrated circuit, and one or more application-specific integrated circuits (ASIC's). Memory **322** may include one or more memory storage locations, and may be in the form of ROM, RAM, flash or other memory types.

(40) Memory **322** is arranged to be accessible to processor **321**, and to contain data **323** that processor **321** is configured to read and write to. Data **323** may store data such as user account data, image data, and data relating to image editing tools, such as machine learning models trained to perform image editing functions.

(41) Memory **322** further comprises program code **324** that is executable by processor **321**, to cause processor **321** to execute workflows. For example, program code **324** comprises a server application **325** executable by processor **321** to cause server system **320** to perform server-side functions. According to some embodiments, such as where image editing application **315** is a web browser, server application **325** may comprise a web server such as Apache, IIS, NGINX, GWS, or an alternative web server. In some embodiments, the server application **325** may comprise an application server configured specifically to interact with image editing application **315**. Server system **320** may be provided with both web server and application server modules.

(42) Program code **324** may also comprise one or more code modules, such as one or more of a background separation module **318** and an inpainting module **319**. As described in further detail below, executing background separation module **318** may cause processor **321** to perform a background separation process, to separate a background and a foreground of an input image. For example, this may be done with a machine learning model, or a pipeline consisting of a combination of machine learning models and traditional image processing techniques. The background separation module may output an alpha channel, as well as color information for the separated foreground elements, in regions where the foreground is not fully opaque. Executing

inpainting module **319** may cause processor **321** to perform an inpainting process. For example, processor **321** may be caused to perform a machine learning or AI based inpainting method, such as LaMa, CoModGANs or diffusion models, such as the publicly available Stable Diffusion model. According to some embodiments, processor **321** may be caused to perform an algorithm based inpainting process, such as PatchMatch.

(43) Background separation module **318** and/or inpainting module **319** may be software modules such as add-ons or plug-ins that operate in conjunction with the server application **325** to expand the functionality thereof. In alternative embodiments, modules **318** and/or **319** may be native to the server application **325**. In still further alternative embodiments, modules **318** and/or **319** may be stand-alone applications (running on server system **320**, or an alternative server system) which communicate with the server application **325**.

(44) While modules **318** and **319** have been described and illustrated as being part of/installed at the server system **320**, the functionality provided by modules **318** and **319** could alternatively be provided by user computing device **310**, for example as an add-on or extension to image editing application **315**, a separate, stand-alone server application that communicates with image editing application **315**, or a native part of image editing application **315**.

(45) Server system **320** also comprises a communications interface **327**, to facilitate communication between server system **320** and other remote or external devices. Communications module **327** may allow for wired or wireless communication between server system **320** and external devices, and may utilise Wi-Fi, USB, Bluetooth, or other communications protocols. According to some embodiments, communications module **327** may facilitate communication between server system **320** and user computing device **310**, for example.

(46) Server system **320** may include additional functional components to those illustrated and described, such as one or more firewalls (and/or other network security components), load balancers (for managing access to the server application **325**), and or other components.

(47) FIG. **4** is a process flow diagram of a method **400** of performing inpainting according to some embodiments. In some embodiments, method **400** may be performed at least partially by processor **311** executing image editing application **315**. In some embodiments, method **400** may be performed at least partially by processor **321** executing server application **325**. While certain steps of method **400** have been described as being executed by particular elements of system **300**, such as by processor **311** or processor **321**, these steps may be performed by different elements in some embodiments.

(48) At step **405**, processor **311** executing image editing application **315** receives an image selected by a user for editing. This may be a result of the user using a camera forming part of the User I/O **316** to capture an image for editing, or by the user selecting an image from a memory location. The memory location may be within the data **313** stored in memory **312** locally on user computing device **310**, or in the data **323** in memory **322** stored remotely in server system **320**. Depending on where the image editing processes are to be performed, a copy of the retrieved image may be stored to a second memory location to allow for efficient access of the image file by processor **311** and/or processor **321**. According to some embodiments, the selected image may be displayed within a user interface of the image editing application **315**, which may be displayed on a display screen forming part of the user I/O **316**. The image editing application **315** may display a number of editing tools for selection by a user to perform image editing functions.

(49) Example images that may be received at step **405** are shown in FIGS. **1**A and **2**A, as described above, and in FIGS. **5**A and **6**A, as described in further detail below.

(50) At step **410**, processor **311** executing image editing application **315** may receive an indication that an inpainting tool has been selected by a user to perform an inpainting process on the received image. According to some embodiments, the selection of the inpainting tool may cause processor **311** to change a cursor or selection method to a brush style tool, allowing a user to brush or paint over the area or areas of the image that they would like removed. In some embodiments, the

selection of the inpainting tool may cause processor **311** to change a cursor or selection method to a trace style tool, allowing a user to trace a perimeter of or circle the area or areas of the image that they would like removed.

(51) At step **415**, processor **311** executing image editing application **315** receives user input corresponding to a selection of the area of the image that they would like removed using the selected inpainting tool, referred to as the user selected area.

(52) An example image showing a user selected area that may be received at step **415** is shown in FIGS. **1**B and **2**B, as described above, and in FIG. **5**B, as described in further detail below.

(53) At step **420**, processor **311** executing image editing application **315** is caused to process the image to separate the background of the image from the foreground of the image. In some embodiments, processor **311** performs step **420** by communicating the image to be processed to server system **320**, and causing processor **321** to execute background separation module **318**. According to some embodiments, this may cause processor **321** to perform a background removal process on the image. According to some embodiments, this may cause processor **321** to perform a foreground removal process on the image. Processor **321** may further be caused to generate a foreground mask and/or a background mask based on the outcome of the separation process, in some embodiments.

(54) An example image having had a background removal process applied is shown in FIG. **6**B, as described in further detail below. An example image showing a foreground mask is shown in FIG. **5**C and FIG. **6**C, as described in further detail below.

(55) At step **425**, processor **311** executing image editing application **315** is caused to determine the amount of overlap between the user selected area as received at step **415** and the foreground and/or background as identified at step **420**. According to some embodiments, processor **311** may calculate a relative mask overlap (RMO) of the user selected area (US_area) with the foreground area (FG_area) based on the following equation:

(56) $RMO = \frac{\text{Num\_pixels(US\_area .Math. FG\_area)}}{\text{Num\_pixels(US\_area)}}$

(57) The numerator of the fraction computes the intersection of the pixels of US_area and FG_area. The denominator computes the number of pixels of US_area. Additionally or alternatively, processor **311** may calculate a relative mask overlap (RMO) of the user selected area (US_area) with the background area (BG_area).

(58) An example image illustrating an overlap between a user selected area and a foreground area is shown in FIGS. **5**D and **7**D, as described in further detail below.

(59) At step **430**, processor **311** executing image editing application **315** is caused to compare the determined value of RMO with one or more threshold values to determine which parts of the image the user intended to inpaint. According to some embodiments, processor **311** may compare the RMO to a lower threshold. The lower threshold may be 0% in some embodiments. According to some embodiments, processor **311** may compare the RMO to an upper threshold. The upper threshold may be between 20% and 90%. In some embodiments, the upper threshold may be between 40% and 60%. For example, the upper threshold may be 50% in some embodiments. In some embodiments, the upper threshold may be calculated by processor **311** based on the size of the user selected area, or the size of the user selected area compared to the accuracy of the selection tool used by the user to select the user selected area. Where the user selected area is small, especially when compared to the tool size, the accuracy of selection is likely to be lower, and the upper threshold value may be selected to be a higher percentage, as it is harder for the user to accurately select a small area of the image for selection. In some embodiments, the threshold values may simply be retrieved from data **313** or data **323**.

(60) Where the RMO is less than or equal to the lower threshold, this indicates that the user selected area is wholly in the background area of the image, and that the user intended to remove an area of the background. Processor **311** executing image editing application **315** proceeds to step **435** to perform an inpainting process on the image, which may be performed by processor **311**

communicating the image to be inpainted to server system **320**, and causing processor **321** to execute inpainting module **319**. This may cause processor **321** to perform an inpainting process on the user selected area of the image.
(61) Where the RMO is more than or equal to the upper threshold, this indicated that the user selected area is largely in the foreground area of the image, and that the user intended to remove an area of the foreground. According to some embodiments, processor **311** may determine whether the foreground area is wholly encompassed in the user selected area. If so, the user may be intending to remove the foreground area, and processor **311** executing image editing application **315** may proceed to step **435** to perform or cause server system **320** to perform an inpainting process on the user selected area of the image. Alternatively, where the user selected area is largely on the foreground area but does not cover the whole foreground area, this may indicate that the user intended to inpaint a portion of the foreground area, and processor **311** may proceed to perform or cause server system **320** to perform an inpainting process on only the foreground area, before combining the inpainted foreground area with the original background area.
(62) Where the RMO is more than the lower threshold but less than the upper threshold, this may indicate that the user wanted to remove something in the background area of the image but happened to brush or trace slightly over the foreground area. In this case, processor **311** may activate a foreground protection algorithm as described below with reference to steps **440** to **470**.
(63) At step **440**, processor **311** executing image editing application **315** is caused to compute a union of the user selected area as received at step **415**, and the foreground area as determined at step **420**. A new mask is created from the results of this union, referred to as the merged mask.
(64) An example image illustrating a union of the user selected area and the foreground area forming a merged mask is shown in FIG. **5**E and FIG. **8**C, as described in further detail below.
(65) At optional step **445**, processor **311** executing image editing application **315** is caused to dilate the merged mask, to ensure any image elements that form part of the foreground area or user selected area have been sufficiently captured. According to some embodiments, the merged mask may be dilated by a predetermined amount, such as a predetermined number of pixels, which may be retrieved from a memory location, such as from data **313** or data **323**. For example, the merged mask may be dilated by between 1 and 10 pixels. In some embodiments, the merged mask may be dilated by between 3 and 7 pixels. In some embodiments, the merged mask may be dilated by 5 pixels. In some alternative embodiments, the number of pixels to dilate the image may be calculated based on the size of the image, and may be a percentage of the number of pixels in the image.
(66) An example image illustrating a dilated merged mask is shown in FIG. **5**F and FIG. **8**D, as described in further detail below.
(67) At step **450**, processor **311** executing image editing application **315** is caused to inpaint the area of the original image as received at step **405** that is spatially aligned with the dilated merged mask determined at step **445** (or the merged mask as determined at step **440**). Step **450** may be performed by processor **311** communicating the image to be inpainted to server system **320**, and causing processor **321** to execute inpainting module **319**. This may cause processor **321** to perform an inpainting process on the masked area. The result is a relatively realistic filling of the masked area. In an ideal case, the image looks realistic and as if the masked area of the image had never been there.
(68) An example image illustrating an inpainting of the merged mask is shown in FIG. **5**G and FIG. **9**C, as described in further detail below.
(69) At optional step **455**, processor **311** executing image editing application **315** is caused to perform a second inpainting step, by inpainting the area of the inpainted image as generated at step **450** that correlates with the user selected area as received at step **415**. As the user selected area is the area of the image from which the unwanted image element is being removed, performing step **455** may result in a more high-resolution and realistic fill of that area. This is because inpainting

processes often produce more high-resolution and realistic results when the area being inpainted is smaller in size, compared to inpainting a larger area. Step **455** may be performed by processor **311** communicating the image to be inpainted to server system **320**, and causing processor **321** to execute inpainting module **319**. This may cause processor **321** to perform an inpainting process on the masked area.
(70) At step **460**, processor **311** executing image editing application **315** is caused to generate a reduced mask, being the user selected area received at step **415** reduced by the foreground area as calculated at step **420**. In other words, the parts of the user selected area that overlap with the foreground area are removed from the user selected area to form the reduced mask.
(71) An example image illustrating a reduced mask is shown in FIG. **5**I and FIG. **11**D, as described in further detail below.
(72) At step **465**, processor **311** executing image editing application **315** is caused to combine the original image as received at step **405** with the inpainted image generated at step **450** or **455** using the reduced mask. Specifically, the areas of the original image corresponding to the reduced mask are replaced with the inpainted image. This causes the background areas of the original image that were selected by the user at step **415** to be filled with the inpainted image, while the foreground image remains untouched.
(73) An example image illustrating the combined image is shown in FIG. **5**K and FIG. **12**D, as described in further detail below.
(74) At optional step **470**, processor **311** executing image editing application **315** is caused to perform a smoothing process to the combined image generated at step **465**. A more realistic result may be achieved if the transition between the original image and the inpainted image is faded in at the edge of the reduced mask. This may be achieved by performing alpha compositing of the foreground area that overlaps the user selected area, such that a few pixels at the boundary of the mask are taken from the foreground area and added to the combined image as generated at step **465**. This may produce a smoother transition. While steps **465** and **470** are illustrated in separate blocks, according to some embodiments, smoothing may be performed at the time the image is combined at step **465**, rather than as an additional step **470**.
(75) An example image illustrating an alpha compositing between the combined image and the foreground area is shown in FIG. **13**D, as described in further detail below.
(76) FIGS. **5**A to **5**K are diagrammatical representations of the stages of performing method **400** on an image.
(77) FIG. **5**A shows an example image **500**, which may be an image received by processor **311** executing image editing application **315** at step **405** of method **400**, as described above. Image **500** includes a foreground element **501**, being a person that is the subject of the captured image, and appears closest to the viewer. Image **500** includes a background element **502**, including a brick wall and some other people. Image **500** also includes an unwanted image element **503**, being a person who appears behind the foreground element **501** and is not the intended subject of the image.
(78) FIG. **5**B shows an example image **505**, which may be an image displayed to a user via user I/O **316** after execution of step **415** of method **400**, as described above. As step **415**, the user selects an area of the original image **500** that they would like removed, which is displayed as user selected area **506**, along with the previously displayed elements of image **500** such as foreground element **501** and background element **502**. In this case, user selected area **506** overlaps unwanted image element **503**. However, user selected area **506** also partially overlaps foreground image element **501**.
(79) FIG. **5**C shows an example image **510**, showing the results of processor **311** executing image editing application **315** performing step **420**, as described above. At step **420**, a background separation step is performed, such that the original image **500** is separated into a foreground area **511** and a background area **512**.
(80) FIG. **5**D shows an example image **515**, showing the overlap of foreground area **511** with user

selected area **506** that may be calculated by processor **311** executing image editing application **315** to perform step **425**, as described above. As described in further detail above, at step **425** processor **311** is caused to calculate the overlap between foreground area **511** and user selected area **506**, to determine the type of inpainting process to perform on the image **500**. With respect to the illustrated embodiment, it is assumed that the amount of overlap is between the lower and upper threshold amount, meaning that the processor **311** determines that a foreground protection algorithm should be performed, as described with respect to steps **440** to **470** above.

(81) FIG. **5**E shows an example image **520**, showing the results of processor **311** executing image editing application **315** to perform step **440**, as described above. At step **440**, processor **311** generates a merged mask **521**, being the union of foreground area **511** and user selected area **506**, as shown in FIG. **5**D.

(82) FIG. **5**F shows an example image **525**, showing the results of processor **311** executing image editing application **315** to perform step **445**, as described above. At step **445**, processor **311** generates a dilated merged mask **526**, by enlarging or dilating the merged mask as shown in FIG. **5**E.

(83) FIG. **5**G shows an example image **530**, showing the results of processor **311** executing image editing application **315** to perform step **450**, as described above. At step **450**, processor **311** inpaints the original image **500** in the area indicated by dilated merged mask **526**. The resulting image **530** includes the background element **502** as in the original image **500**. However, the foreground element **501** and the unwanted image element **503** have been inpainted to produce inpainted area **531**. In other words, foreground element **501** and the unwanted image element **503** have been replaced or filled with background imagery generated by processor **311**, to create a realistic image that causes the impression that foreground element **501** and the unwanted image element **503** were never in the image.

(84) FIG. **5**H shows an example image **535**, showing the intersection between foreground area **511** and user selected area **506**, which may be used by processor **311** executing image editing application **315** to perform step **460**, as described above. At step **460**, processor **311** generates a reduced mask, being the user selected area **506** reduced by the foreground area **511**.

(85) FIG. **5**I shows an example image **540**, showing the results of processor **311** executing image editing application **315** to perform step **460**, as described above. At step **460**, processor **311** generates a reduced mask **536**, by subtracting the foreground area **511** from the user selected area **506** as shown in FIG. **5**H.

(86) FIG. **5**J shows an example image **545**, showing the reduced mask **536** overlaid on the inpainted image **530**, which may be used by processor **311** executing image editing application **315** to perform step **465**, as described above. At step **465**, processor **311** combines original image **500** with the inpainted image **530** in the area defined by reduced mask **536**. In some embodiments, this may be done by copying the area of image **540** that overlaps with reduced mask **536**, and pasting this onto the same area of the original image **500**.

(87) FIG. **5**K shows an example image **550**, showing the results of processor **311** executing image editing application **315** to perform step **465**, as described above. At step **465**, processor **311** combined original image **500** with inpainted image **530** to produce a final image **545**, where the unwanted image element **503** has been removed to produce inpainted area **546**, while preserving foreground element **501**.

(88) FIGS. **6**A to **6**C show a series of example images showing the results of processor **311** executing image editing application **315** to perform step **420**, being a background separation step, as described above.

(89) FIG. **6**A shows an example image **600**, which may be an image received by processor **311** executing image editing application **315** at step **405** of method **400**, as described above. Image **600** includes a foreground element **605**, being a person that is the subject of the captured image, and appears closest to the viewer. Image **600** includes a background element **610**, including a floor and

other people. Image **600** also includes an unwanted image element **615**, being a person who appears behind the foreground element **605** and is not the intended subject of the image.
(90) FIG. **6**B shows an example image **630** illustrating the results of processor **311** executing image editing application **315** to perform a background removal step on image **600**, which may be done by processor **311** communicating the image to be processed to server system **320**, and causing processor **321** to execute background separation module **318**. In image **630**, foreground element **605** has been retained, but background element **610** including unwanted image element **615** has been removed, leaving just a background area **635**.
(91) FIG. **6**C shows an example image **660** illustrating the results of processor **311** executing image editing application **315** to perform a foreground removal step on image **630**, which may be done by processor **311** communicating the image to be processed to server system **320**, and causing processor **321** to execute background separation module **318**. In image **660**, foreground element **605** has been removed, leaving just a foreground area **665** and a background area **670**. Foreground area **665** and background area **670** may be used as masks by processor **311** in further processing steps.
(92) FIGS. **7**A to **7**D show a series of example images showing the results of processor **311** executing image editing application **315** to perform step **425**, being a relative mask overlap calculation step, as described above.
(93) FIG. **7**A shows the original image **600**, as described above.
(94) FIG. **7**B shows an example image **700** having a user selected area **710** which was selected by a user of user computing device **310** as an area of the original image **600** that they would like removed. The user selected area **710** corresponds with the unwanted image element **615**. However, user selected area **710** also overlaps foreground element **605**.
(95) FIG. **7**C shows an example image **660** separated into a foreground area **665** and a background area **670**, as described above.
(96) FIG. **7**D shows an image **750** showing the user selected area **710** of FIG. **7**B overlaid on the foreground area **665** of FIG. **7**C. This image may be used by processor **311** to perform step **425**, to determine the amount of overlap **760** between user selected area **710** and foreground area **665**.
(97) FIGS. **8**A to **8**D show a series of example images showing the results of processor **311** executing image editing application **315** to perform steps **440** and **445**, being a merged mask generation step and a dilated merged mask generation step, as described above.
(98) FIG. **8**A shows image **700** illustrating the user selected area **710**, as described above.
(99) FIG. **8**B shows the separated image **660** separated into a foreground area **665** and a background area **670**, as described above.
(100) FIG. **8**C shows an image **800** illustrating a merged mask **810** generated by processor **311** executing image editing application **315** to perform step **440**, as described above. At step **440**, processor **311** calculates the union of the user selected area **710** as shown in FIG. **8**A and the foreground area **665** as shown in FIG. **8**B to produce merged mask **810**.
(101) FIG. **8**D shows an image **850** illustrating a dilated merged mask **860** generated by processor **311** executing image editing application **315** to perform step **440**, as described above. At step **445**, processor **311** dilates the merged mask **810** as shown in FIG. **8**C to produce dilated merged mask **860**.
(102) FIGS. **9**A to **9**C show a series of example images showing the results of processor **311** executing image editing application **315** to perform step **450**, being a merged mask inpainting step, as described above.
(103) FIG. **9**A shows the original image **600**, as described above.
(104) FIG. **9**B shows image **850** illustrating a dilated merged mask **860**, as described above.
(105) FIG. **9**C shows an image **900** generated by processor **311** executing image editing application **315** to perform step **450**, as described above. At step **450**, processor **311** performs inpainting of image **600** as shown in FIG. **9**A in the area indicated by the dilated merged mask **860** as shown in

FIG. **9**B. This produces inpainted image **900**, in which the foreground element **605** and unwanted image element **615** have been inpainted to produce an inpainted area **910**. Background element **610** is otherwise untouched.
(106) FIGS. **10**A to **10**C show a series of example images showing the results of processor **311** executing image editing application **315** to perform step **455**, being a user selected mask inpainting step, as described above.
(107) FIG. **10**A shows the inpainted image **900**, as described above.
(108) FIG. **10**B shows image **700** illustrating user selected area **710**, as described above.
(109) FIG. **10**C shows an image **1000** generated by processor **311** executing image editing application **315** to perform step **455**, as described above. At step **455**, processor **311** performs inpainting of image **900** as shown in FIG. **10**A in the area indicated by user selected area **710** as shown in FIG. **10**B. This produces a further inpainted image **1000**, in which the area corresponding to user selected area **710** has been inpainted again to produce a further inpainted area **1010**. As described above, this second inpainting step may produce a more high resolution and/or realistic result in the area of the image corresponding to the user selected area **710**.
(110) FIGS. **11**A to **11**D show a series of example images showing the results of processor **311** executing image editing application **315** to perform step **460**, being a reduced mask generation step, as described above.
(111) FIG. **11**A shows the original image **600**, as described above.
(112) FIG. **11**B shows image **700** illustrating user selected area **710**, as described above.
(113) FIG. **11**C shows the separated image **660** separated into a foreground area **665** and a background area **670**, as described above.
(114) FIG. **11**D shows an image **1100** generated by processor **311** executing image editing application **315** to perform step **460**, as described above. At step **460**, processor **311** generates a reduced mask **1110** by subtracting foreground area **665** as shown in FIG. **11**C from user selected area **710** as show in FIG. **11**B.
(115) FIGS. **12**A to **12**D show a series of example images showing the results of processor **311** executing image editing application **315** to perform step **465**, being a reduced mask inpainting step, as described above.
(116) FIG. **12**A shows the original image **600**, as described above.
(117) FIG. **12**B shows image **1100** illustrating the reduced mask **1110**, as described above.
(118) FIG. **12**C shows the inpainted image **1000**, as described above.
(119) FIG. **12**D shows an image **1200** generated by processor **311** executing image editing application **315** to perform step **465**, as described above. At step **465**, processor **311** combines the original image **600** as shown in FIG. **12**A with the inpainted image **1000** as shown in FIG. **12**C using the reduced mask **1110** as shown in FIG. **12**B. Specifically, the area of the original image **600** corresponding to the user selected area **710** is replaced by the corresponding area of inpainted image **1000**. Resulting image **1200** includes the background element **610** and foreground element **605** of image **600**, but unwanted image element **615** has been replaced by inpainted area **1010** from image **1000**.
(120) FIGS. **13**A to **13**D show a series of example images showing the results of processor **311** executing image editing application **315** to perform step **470**, being a smooth image merging step, as described above.
(121) FIG. **13**A shows the original image **600**, as described above.
(122) FIG. **13**B shows the separated image **660** separated into a foreground area **665** and a background area **670**, as described above.
(123) FIG. **13**C shows the combined image **1200**, as described above.
(124) FIG. **13**D shows an image **1300** generated by processor **311** executing image editing application **315** to perform step **470**, as described above. At step **470**, processor **311** performs a smoothing process on the combined image **1200** as shown in FIG. **13**C by executing an alpha

compositing of the foreground element **605** of FIG. **13**A that overlaps with user selected area **710**. By adding pixels from foreground element **605** of FIG. **13**A to combined image **13**C in a graduating manner as the image transitions from foreground area **665** and background area **670**, a smoother and more realistic image can be produced. Resulting image **1300** has a smoother and more realistic transition **1305** between foreground element **605** and inpainted area **1010**. As described above with reference to FIG. **4**, the smoothing process performed at **470** may be performed simultaneously with the process of combining the original image **600** with the inpainted image **1000**. As a result, when the smoothing process is performed, image **1300** may be generated directly by combining original image **600** with the inpainted image **1000** using an alpha compositing process, instead of first generating image **1200**.

(125) It will be appreciated by persons skilled in the art that numerous variations and/or modifications may be made to the above-described embodiments, without departing from the broad general scope of the present disclosure. The present embodiments are, therefore, to be considered in all respects as illustrative and not restrictive.

Claims

1. A computer-implemented method for performing inpainting, the method comprising: accessing a first image; receiving a selected area of the first image; identifying a foreground area of the first image; generating a merged mask that covers the entirety of a union of the selected area and the foreground area; performing an inpainting process on an area of the first image corresponding to the merged mask to generate a second image, being an inpainted image; generating a reduced mask based on the selected area of the first image reduced by the foreground area of the first image; and combining the first image with an area of the second image corresponding to the reduced mask to produce an output image, wherein areas of the first image corresponding to the reduced mask are replaced with the inpainted image.

2. The method of claim 1, further comprising storing the output image to a memory location and/or communicating the output image to an external device.

3. The method of claim 1, further comprising: computing a relative mask overlap between the selected area and the foreground area; comparing the relative mask overlap with at least one threshold; and proceeding with the steps of generating a merged mask, performing an inpainting process, generating a reduced mask and combining images based on results of the comparison, wherein these steps form part of a foreground protection process.

4. The method of claim 3, wherein the relative mask overlap is calculated using the equation: $\text{RMO} = \frac{\text{Num\_pixels(US\_area .Math. FG\_area)}}{\text{Num\_pixels(US\_area)}}$ where RMO is the relative mask overlap, Num_pixels is a function to find a number of pixels, US_area is the selected area and FG_area is the foreground area.

5. The method of claim 3, wherein comparing the relative mask overlap with at least one threshold comprises comparing the relative mask overlap with a lower threshold, and performing the foreground protection process based on determining that the relative mask overlap is greater than the lower threshold.

6. The method of claim 5, wherein the lower threshold is 0%.

7. The method of claim 5, wherein based on determining that the relative mask overlap is less than or equal to the lower threshold, instead of performing the foreground protection process, performing an inpainting process on the area of the first image corresponding to the selected area to generate the output image.

8. The method of claim 3, wherein comparing the relative mask overlap with at least one threshold comprises comparing the relative mask overlap with an upper threshold, and performing the foreground protection process based on determining that the relative mask overlap is less than the

upper threshold.
9. The method of claim 8, wherein the upper threshold is: between 10% and 90%; or between 40% and 70%; or 50%.
10. The method of claim 8, wherein based on determining that the relative mask overlap is more than or equal to the upper threshold, instead of performing the foreground protection process, performing an inpainting process on the area of the first image corresponding to the selected area to generate the output image.
11. The method of claim 1, further comprising the step of dilating the merged mask, and performing the inpainting process on the area of the first image corresponding to the dilated merged mask.
12. The method of claim 11, wherein the merged mask is dilated by: between 1 and 9 pixels; or between 3 and 7 pixels; or 5 pixels.
13. The method of claim 1, further comprising performing a second inpainting process on the area of the first image corresponding to the selected area to generate the second image.
14. The method of claim 1, further comprising performing a smoothing process on the output image before outputting the image, to smooth the edge between the first image and the second image.
15. The method of claim 14, wherein the smoothing is performed using an alpha compositing process.
16. The method of claim 1, wherein performing the inpainting process comprises performing an AI based inpainting process.
17. The method of claim 16, wherein performing the inpainting process comprises performing a LaMa inpainting process.
18. The method of claim 1, wherein identifying a foreground area of the first image comprises performing a background removal process on the first image.
19. Non-transitory computer-readable storage medium storing instructions which, when executed by a processing device, cause the processing device to perform a method comprising: accessing a first image; receiving a selected area of the first image; identifying a foreground area of the first image; generating a merged mask that covers the entirety of a union of the selected area and the foreground area; performing an inpainting process on an area of the first image corresponding to the merged mask to generate a second image, being an inpainted image; generating a reduced mask based on the selected area of the first image reduced by the foreground area of the first image; and combining the first image with an area of the second image corresponding to the reduced mask to produce an output image, wherein areas of the first image corresponding to the reduced mask are replaced with the inpainted image.
20. A computing device comprising: the non-transitory computer-readable storage medium of claim 19; and a processor configured to execute the instructions stored in the non-transitory computer-readable storage medium.